INVERTIS UNIVERSITY BAREILLY
Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

BCS-071	Embedded and Real Time Systems	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	Syllabus deals with issues in real time operating systems
CO2	Importance of deadlines and concept of task scheduling
CO3	Student will be able to understand and design real time operating systems which are backbone of embedded industry
CO4	Student will be able to understand Common Approaches to Real Time Scheduling
CO5	To learn about RAC

Detailed Syllabus

MODULE-I

Introduction: Embedded Systems and general purpose computer systems, history, classifications, applications and purpose of embedded systems. Core of Embedded Systems: Microprocessors and microcontrollers, RISC and CISC controllers, Big endian and Little endian processors, Application specific ICs, Programmable logic devices, COTS, sensors and actuators, communication interface, embedded firmware, other system components, PCB and passive components.

MODULE-II

Introduction: Definition, Typical Real Time Applications: Digital Control, High Level Controls, Signal Processing etc., Release Times, Deadlines, and Timing Constraints, Hard Real Time Systems and Soft Real Time Systems, Reference Models for Real Time Systems: Processors and Resources, Temporal Parameters of Real Time Workload, Periodic Task Model, Precedence Constraints and Data Dependency.

MODULE-III

Real Time Scheduling: Common Approaches to Real Time Scheduling: Clock Driven Approach, Weighted Round Robin Approach, Priority Driven Approach, Dynamic Versus Static Systems, Optimality of Effective-Deadline-First (EDF) and Least-Slack-Time-First (LST) Algorithms, Rate Monotonic Algorithm, Offline Versus Online Scheduling, Scheduling Aperiodic and Sporadic jobs in Priority Driven and Clock Driven Systems.

Resources Sharing: Effect of Resource Contention and Resource Access Control (RAC), Non-preemptive Critical Sections, Basic Priority-Inheritance and Priority-Ceiling Protocols, Stack Based Priority- Ceiling Protocol, Use of Priority-Ceiling Protocol in Dynamic Priority Systems, Preemption Ceiling Protocol.

Course Outcomes: After the completion of the course the student will be able to:

CO1	To present the mathematical model of the system.
CO2	To develop real-time algorithm for task scheduling.
CO3	To understand the working of real-time operating systems and real-time database
CO4	To work on design and development of protocols related to real-time communication.
CO5	Able to implement RAC

Text Books:-

1. Real Time Systems by Jane W. S. Liu, Pearson Education Publication.
2. Embedded Systems, Rajkamal, TataMcGraw-Hill

Reference Books:

1. Mall Rajib, "Real Time Systems", Pearson Education
2. Albert M. K. Cheng , "Real-Time Systems: Scheduling, Analysis, and Verification",Wiley.